

Your grade: 100%

Next item →

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1.

When implementing ResNet for image prediction, which of the following steps are typically involved?

1 / 1 point

☒ Fine-tuning the model on the target dataset

✔ Correct

Correct. Fine-tuning allows the model to adapt to the specific characteristics of the target dataset.

☒ Choosing an appropriate ResNet architecture

✔ Correct

Correct. Selecting the right ResNet variant (e.g., ResNet-50, ResNet-101) is essential based on the complexity of the task.

☐ Training the model from scratch

☐ Freezing all the layers of ResNet

☒ Preprocessing the input images

✔ Correct

Correct. Preprocessing is crucial for normalizing the input data and improving model performance.

2.

Which steps are involved in customizing the VGG16 model for image classification?

1 / 1 point

☒ Removing dense layers from the original VGG16 model.

✔ Correct

Correct! Removing dense layers is one of the steps involved in customizing the VGG16 model.

☒ Specifying the input shape to match the new dataset.

✔ Correct

Correct! Specifying the input shape is a crucial step in customization.

☐ Training the modified model on the new dataset from scratch.

☒ Adding new dense layers suitable for the specific classification task.

✔ Correct

Correct! Adding new dense layers suitable for the new task is essential.

☐ Using the original VGG16 model without any modifications.

3.

What is the significance of specifying the input shape when modifying the VGG16 model for a specific dataset?

1 / 1 point

☐ It reduces the number of parameters in the model.

☒ It ensures that the input data dimensions match the expected dimensions of the model.

☐ It allows the model to handle different types of data formats.

☐ It helps in normalizing the input data.

✔ Correct

Correct! Specifying the input shape ensures that the input data dimensions match the model's expected dimensions.

4.

Which of the following pre-trained models can be used for transfer learning in image classification tasks?

1 / 1 point

☐ DeepLab

☒ InceptionV3

✔ Correct

Correct! InceptionV3 is commonly used for transfer learning.

☒ ResNet50

✔ Correct

Correct! ResNet50 is also widely used for transfer learning.

☒ SqueezeNet

✔ Correct

Correct! SqueezeNet can be used for transfer learning.

☒ MobileNet

✔ Correct

Correct! MobileNet is another popular choice for transfer learning.

5.

What is the primary benefit of using transfer learning with the VGG16 model for image classification tasks?

1 / 1 point

☒ Reduced training time

☐ Automatic feature extraction

☐ Improved accuracy without any modifications

☐ Ability to handle larger datasets

✔ Correct

Correct! Transfer learning significantly reduces training time compared to training from scratch.

https://www.coursera.org/learn/packt-deep-learning-with-keras-and-practical-applications-vcwvs/assignment-submission/38orP/assessment-10/view-feedback

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